Arista Access Node HEB-ACC-25 Optical Transport Failure (ALARM_5885) SOP

Node: HEB-ACC-25 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5885

Description

The Optical Transport Failure (ALARM_5885) in Arista Access Node HEB-ACC-25 at Hebbal Edge is primarily caused by SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. This failure domain occurs on the Access role due to the reliance on optical transport mechanisms for data transmission. Arista's EoS (End-of-Sales) policy for specific SFP/XFP modules, along with the inherent fragility of optical fibers, contributes to the likelihood of such failures. Moreover, the complexity of the optical transport network, comprising multiple fibers, amplifiers, and transceivers, increases the probability of failure. In this SOP, we will outline the steps to diagnose and resolve the Optical Transport Failure (ALARM_5885) in Arista Access Node HEB-ACC-25.

Pre-requisite Checks

- check: Verify that the Hebbal Edge site is accessible to remote hands and that all necessary equipment is on site.

- check: Confirm that the redundant link status is operational to ensure minimal service impact during the resolution process.

- check: Ensure that all necessary personnel are aware of the potential risks associated with the resolution process.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface status and operational details, Description: This command will provide an overview of the interface status, including the operational state and any alarm conditions.

Command: show interface transceiver, Expected Output: Transceiver details, including temperature, voltage, and optical parameters, Description: This command will provide detailed information about the transceiver, including temperature, voltage, and optical parameters, which can help identify potential issues.
```

```
Command: show alarm, Expected Output: Alarm details, including
severity and description, Description: This command will provide alarm
details, including severity and description, which can help identify
the root cause of the failure.
```

Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP module, Command: install module in slot , Description: Replace the faulty SFP/XFP module with a new one. Ensure that the new module is compatible with the Arista Access Node HEB-ACC-25 and the Hebbal Edge site configuration.

Step 2: Step: Verify fiber integrity, Command: show interface fiber, Description: Verify the fiber integrity by checking for any cuts, damage, or degradation. If any issues are found, replace the fiber accordingly.

Step 3: Step: Adjust Rx/Tx power levels, Command: set interface rx-power tx-power , Description: Adjust the Rx/Tx power levels to optimize the signal strength and quality. Ensure that the new power levels are within the recommended range for the Arista Access Node HEB-ACC-25 and the Hebbal Edge site configuration.

Post-Check Verification

```
Command: show interface status, Expected Output: Interface status and
operational details, Description: Verify that the interface is
operational and that the alarm condition has been cleared.

Command: show alarm, Expected Output: Alarm details, including
severity and description, Description: Verify that the alarm condition
has been cleared and that no new alarm conditions have been raised.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: no install module , Description: Restore the original SFP/XFP module if the replacement module does not resolve the issue.

- Step: Revert fiber changes, Command: set interface fiber , Description: Revert any fiber changes made during the resolution process if they do not resolve the issue.

- Step: Revert Rx/Tx power changes, Command: set interface rx-power tx-power , Description: Revert any Rx/Tx power changes made during the resolution process if they do not resolve the issue.